

CURE: Constrained Uncertainty Routing with Complementary Experts for Retrieved-Evidence Fact Verification

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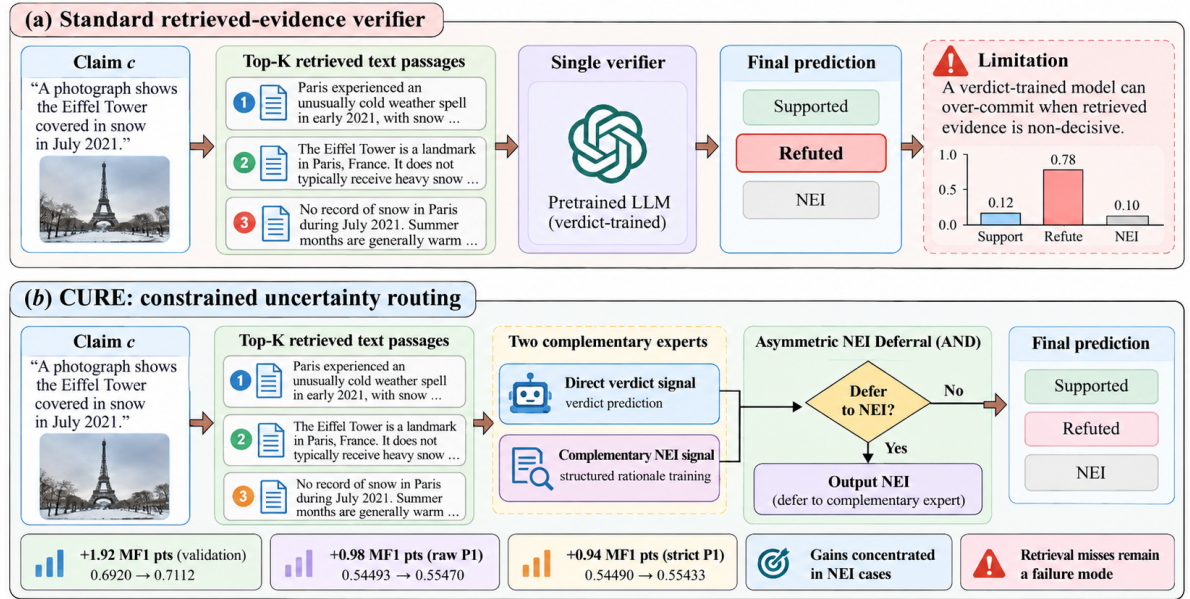


Figure 1: Overview of CURE, a constrained uncertainty routing framework with complementary experts for retrieved-evidence fact verification. **(a)** A standard pipeline uses a single verdict-trained model, which can over-commit when retrieved evidence is non-decisive. **(b)** CURE routes between a direct verdict expert and a complementary NEI expert via Asymmetric NEI Deferral (AND), leading to more reliable predictions. Bottom: key results and takeaways.

Abstract

Fact verifiers often conflate two distinct decisions: which verdict is most likely and whether the retrieved evidence warrants a decisive verdict. We study this distinction in a fixed-corpus, text-evidence MOCHEG protocol. CURE (Constrained Uncertainty Routing with Complementary Experts) combines a verdict-trained direct expert with a rationale-trained complementary expert. Its *Asymmetric NEI Deferral* rule preserves the direct verdict by default and permits only a high-confidence complementary NEI prediction to defer the output. On frozen validation, this constrained rule improves Macro-F1 from 0.6920 to 0.7112 (+1.92 points; paired-bootstrap $P(\Delta > 0) =$

0.9975), whereas equal averaging and symmetric confidence routing do not. With frozen $K = 5$ and $\tau = .49$, the raw/strict P1 test point estimates are +0.00977/+0.00943 MF1 (95% CIs $[-.00128, .02109]/[-.00187, .02065]$; $P(\Delta > 0) = .9529/.9477$). These are directional rather than statistically resolved test results. Matched teacher controls find unresolved 3B-versus-7B capacity differences and near-identical Mistral-7B versus Qwen2.5-7B results, ruling out simple capacity or tested-family explanations. Post-hoc diagnostic groups suggest a benefit for some NEI cases and identify retrieval misses as a failure mode. A frozen zero-shot SciFact audit finds that the MOCHEG policy does not transfer without

adaptation. A separate validation-selected heterogeneous ensemble reaches 0.5551 Macro-F1 on raw P1, and a disagreement-distilled student offers a lower-latency extension.

1 Introduction

Automated fact checking is commonly decomposed into evidence retrieval and claim verification (Guo et al., 2022; Thorne et al., 2018). Modern systems increasingly retrieve heterogeneous evidence and use large language models, but two assumptions remain fragile: that all retrieved passages should be consumed by the verifier, and that a model trained to predict a verdict is also a reliable detector of *retrieved-evidence sufficiency*. We focus on the latter distinction in a controlled text-evidence setting.

MOCHEG (Yao et al., 2023) provides a useful setting because it couples three-way truthfulness labels with textual, visual, and social evidence. We deliberately isolate its text-evidence component: all experts receive the same fixed-corpus retrieved passages, so that effects of supervision and decision policy are not confounded by live-web retrieval or visual modeling. This is therefore a fact-verification study on a multimodal benchmark, rather than a claim that the proposed system itself is multimodal.

Our central observation is that two verifiers have complementary error profiles. A direct verifier is strong when evidence appears decisive but can overcommit under non-decisive retrieved context. A rationale-trained expert is more likely to predict NEI, but is not uniformly more accurate and is not assumed to be a calibrated sufficiency estimator. We turn this complementarity into a constrained policy rather than averaging the experts. The direct expert remains the default; only a high-confidence complementary NEI signal can defer the decision. We call this operator *Asymmetric NEI Deferral* (AND). It is a class-constrained cross-expert decision rule, not Bayesian epistemic uncertainty estimation.

Our contributions are:

- CURE, a dual-expert formulation that combines direct verdict prediction with a rationale-trained complementary NEI signal;
- AND, a class-constrained cross-expert rule that permits only NEI deferral, and controlled ablations against rationale-trained-only, poste-

rior averaging, and symmetric routing alternatives;

- post-hoc diagnostic analyses that identify NEI improvements and retrieval-miss failure cases, without using diagnostic labels as routing inputs; and
- a disagreement-distilled student as a secondary, lower-latency deployment extension.

We distinguish frozen-test evaluation from development diagnostics. The heterogeneous ensemble and AND answer different questions: the former tests broad probability complementarity, while the latter tests a fixed, class-constrained decision policy. Both are evaluated on their respective frozen test protocols without test-time parameter selection.

2 Related Work

Fact verification and evidence sufficiency. FEVER established evidence-supported three-way verification at scale (Thorne et al., 2018); LIAR studied short political claims (Wang, 2017); and MultiFC broadened evaluation across sources and domains (Augenstein et al., 2019). SciFact further couples scientific claim labels with evidence rationales (Wadden et al., 2020). These benchmarks make clear that label prediction, evidence retrieval, and the availability of a convincing rationale are not identical targets. Surveys consequently treat retrieval, evidence selection, reasoning, and calibration as coupled but distinct stages (Guo et al., 2022). UKP-Athene illustrates the role of multi-sentence textual entailment in evidence-based verification (Hanselowski et al., 2018).

Retrieval, selection, and reasoning. Evidence retrieval can dominate downstream verification (Zheng et al., 2024); evidence selection itself materially changes end-to-end performance (Samarinas et al., 2021). RAV combines retrieval with verification, while RAFTS synthesizes contrastive arguments from retrieved evidence (Yue et al., 2024). Complex-claim verification decomposes claims before retrieving web evidence (Chen et al., 2024a). These approaches study different evidence regimes from our P1 protocol: they motivate the separation between fixed-corpus evidence selection and live-web retrieval rather than serving as directly interchangeable leaderboard rows.

Rationales, selective prediction, and expert routing. Extractive and generated rationales have been studied as both explanations and learning signals (Lei et al., 2016; Atanasova et al., 2020); ERASER emphasizes that plausibility and faithfulness require separate evaluation (DeYoung et al., 2020). Our matched-recipe control follows this caution: rationale supervision is not assumed to cause a verdict gain. AND is also related to selective prediction with a reject option (Geifman and El-Yaniv, 2019), but it does not learn a reject head or optimize risk-coverage. Instead, it compares a frozen direct posterior with a complementary expert and permits one class-specific deferral. This differs from sparsely gated mixture-of-experts routing (Shazeer et al., 2017); CURE evaluates all members sequentially and uses a fixed, interpretable decision rule rather than learned conditional computation. Because softmax confidence need not be calibrated (Guo et al., 2017), we report calibration diagnostics and explicitly compare against direct self-deferral.

MOCHEG and multimodal fact checking. MOCHEG introduced end-to-end multimodal fact checking and explanation generation (Yao et al., 2023). Recent systems pursue different ways to aggregate textual and visual evidence: LVLM4FV uses large vision-language models with reranked evidence (Tahmasebi et al., 2024); MetaSumPerceiver creates claim-specific multimodal summaries (Chen et al., 2024b); HGTMF models high-order interactions with hypergraphs (Pang et al., 2025); MEVER jointly addresses retrieval, verification, and explanations (Zhang et al., 2026); and M-RAV augments cross-modal evidence alignment for zero-shot verification (Luu et al., 2027). AMuFC adaptively determines whether visual evidence is needed (Jung et al., 2026). These systems motivate the benchmark comparison, but our P1 method consumes retrieved text only. CURE consequently studies constrained routing between differently trained text verifiers, not multimodal fusion.

3 Task Definition and Protocols

Given a claim c and retrieved evidence $E = \{e_j\}_{j=1}^K$, predict $y \in \{\text{SUPPORTED}, \text{REFUTED}, \text{NEI}\}$. The principal P1 protocol uses MOCHEG’s fixed corpus and upstream text retrieval; it is not an open-web evaluation. The raw official test contains 2,442 claims. A strict robustness track removes eight

verbatim cross-split duplicates and contains 2,434 claims. We report both tracks separately. Development validation contains 1,456 claims.

All routing thresholds are selected on validation or train-only out-of-fold data. Test labels are never used for selecting a seed, threshold, or policy in frozen-test evaluation. Open-web experiments are not mixed into P1 tables.

4 Method

4.1 Direct and rationale-trained experts

The direct expert f_D receives a claim c and the top- K reranked passages E_K and predicts a three-way distribution $p_D(y \mid c, E_K)$. It is optimized only with verdict cross-entropy. The rationale-trained expert f_G has the same Qwen3-4B-LoRA backbone and the same inference prompt, but is additionally trained to generate a structured teacher rationale. Thus, it remains a verdict classifier at inference rather than a rationale generator. Its auxiliary targets are schema-validated against retrieved passages; this provenance check does not assert that $p_G(\text{NEI})$ is a calibrated measure of evidence sufficiency.

The primary rationale targets are generated once for training claims by Qwen2.5-7B-Instruct using the *gold training verdict* and the system-retrieved top-five passages. The teacher is instructed to flag a rationale when those passages cannot justify that training verdict. A valid target contains a retrieval-validation flag, verdict, key-passage identifiers, free-text reason, and missing-information field. Generation is deterministic (greedy decoding, maximum 512 new tokens), and schema validation rejects malformed key-passage references and unsupported structured outputs. Explanation loss is applied only to valid, retrieval-supported targets. Later matched ablations replace only this teacher with Qwen2.5-3B-Instruct or Mistral-7B-Instruct-v0.3. This use of training labels is ordinary supervised learning; no validation or test label, and no gold evidence passage, is included in a validation/test prompt.

For a training example i , let z_i be the verdict target and r_i the structured rationale target. The two expert objectives are

$$\begin{aligned}\mathcal{L}_D &= - \sum_i \log p_D(z_i \mid c_i, E_i), \\ \mathcal{L}_G &= \mathcal{L}_D + \lambda_r \mathcal{L}_{\text{rationale}}(r_i), \quad \lambda_r = 0.25.\end{aligned}\tag{1}$$

The rationale term is token-level cross-entropy over the validated structured target. Both experts begin from the same base model. We report two recipe controls: a five-fold train-only OOF explanation-supervision contrast under a common recipe, and a fresh direct-only control matching the headline rationale-trained data, optimizer, batch size, accumulation, learning rate, epochs, and seeds. The latter reduces the headline recipe confound, although it remains a validation development comparison because both arms select their checkpoints on validation. The implementation records teacher outputs, manifests, retrieval signatures, and prompt-format hashes. Gold evidence is used only for retrieval diagnostics, never injected into P1 validation or test prompts.

4.2 Asymmetric NEI deferral (AND)

The policy uses the direct expert as an anchor:

$$\hat{y} = \begin{cases} \text{NEI}, & \arg \max_y p_G(y) = \text{NEI} \wedge p_G(\text{NEI}) \geq \tau, \\ \arg \max_y p_D(y), & \text{otherwise.} \end{cases} \quad (2)$$

We use $\tau = 0.49$, selected on validation and frozen thereafter for the registered $K = 5$ router. This rule is intentionally asymmetric: the rationale-trained expert is not permitted to overwrite a direct Supported or Refuted decision with another decisive class. It can only defer to NEI when its complementary NEI posterior is high. AND is therefore distinct from posterior averaging, symmetric confidence routing, and an unconstrained expert switch. $p_G(\text{NEI})$ is a learned class posterior, not a Bayesian estimate of parameter uncertainty or a directly validated sufficiency probability.

4.3 Disagreement-aware distillation

CURE-Student is a secondary single-model variant. Let q_D and q_G be frozen teacher distributions generated without the held-out fold. For agreement cases, the distillation target is their average; for disagreement cases, a train-only cross-fitted outcome identifies which teacher target to transfer. The confirmed variant minimizes

$$\mathcal{L}_{\text{DKD}} = \mathcal{L}_{\text{CE}} + \lambda_{\text{KD}} T^2 \text{KL}(q \| p_S). \quad (3)$$

Here q is the selected teacher target and $T = 2$ is the distillation temperature. A counterfactual sufficiency loss was tested during the Fold-0 mechanism screen but was not retained in the confirmed disagreement-KD variant. CURE-Student is reported as a compute-efficient secondary extension, not as a replacement for AND.

Split/protocol	Claims	Labels used for selection	Role
Train	11,631	train only	optimization
Validation	1,456	validation	frozen policy selection
Official test	2,442	no	primary benchmark
Strict test	2,434	no	duplicate-robustness

Table 1: Protocol sizes. Official and strict test results are not pooled.

4.4 Competitive ensemble variant

The CURE-Ensemble test system is a complementary, validation-selected ensemble: five frozen direct seeds plus the three best rationale-trained seeds ranked on validation. The composition $\{100, 87, 42\}$ is fixed before the official test is evaluated, and its prediction is the arithmetic mean of member verdict probabilities. We also report an unpruned ten-member ensemble (all five direct plus all five rationale-trained seeds) as a no-subset-selection control. This system is a competitive benchmark variant, not an implementation of AND: it answers whether heterogeneous expert predictions are complementary, whereas AND answers whether a constrained NEI deferral is useful. To isolate member count, we additionally train three predeclared direct-only seeds $\{314, 2718, 2026\}$ with the legacy direct recipe. The resulting direct-only eight-member control gives every member equal weight and is compared with the five-direct-plus-three-rationale ensemble under the same raw/strict manifests.

5 Experimental Setup

5.1 Data and preprocessing

MOCHEG contains 15,601 English claims from PolitiFact and Snopes with textual, visual, and social evidence (Yao et al., 2023). Our P1 experiments use a fixed MOCHEG *text* corpus. Qwen3-Embedding-4B retrieves a top-50 dense candidate pool, and Qwen3-Reranker-4B then scores claim–passage utility with an instruction requiring entity, temporal, quantity, location, and negation compatibility. The registered router consumes the top five reranked passages ($K = 5$), truncated to 2,200 evidence characters. No gold evidence is injected at validation or test. The open-web C-series snapshot is intentionally excluded from P1 effectiveness tables because it is a different deployment regime.

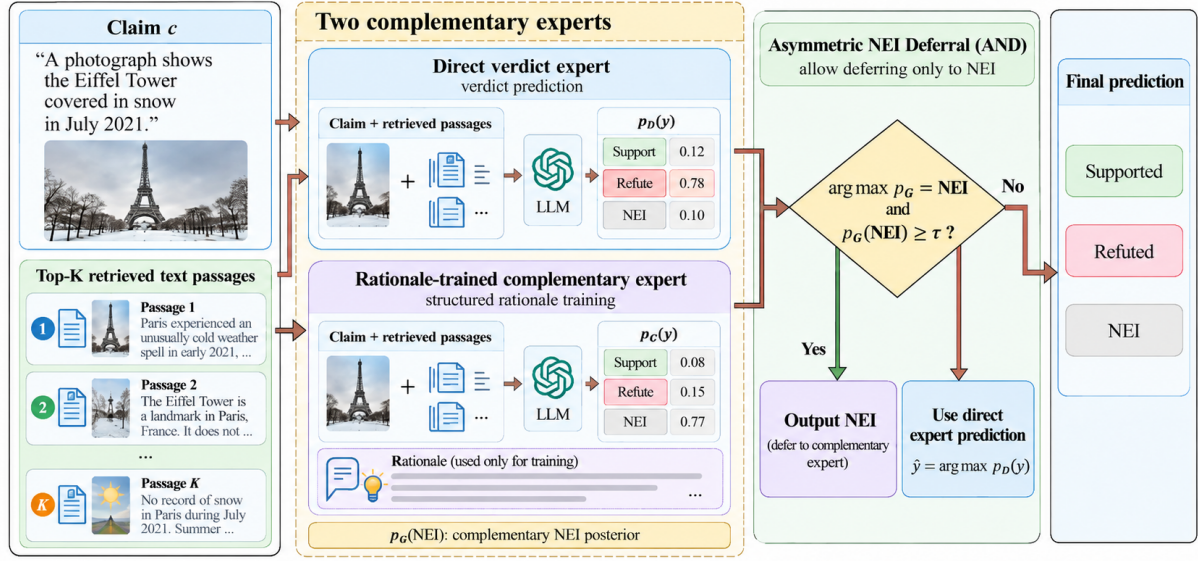


Figure 2: Detailed overview of CURE. The claim and retrieved passages are processed by a direct verdict expert and a rationale-trained complementary expert. AND preserves the direct verdict by default and defers only when the complementary expert predicts NEI with sufficiently high confidence.

5.2 Models and metrics

All verifiers use Qwen3-4B-Instruct-2507 with LoRA adapters. The legacy direct ensemble uses seeds $\{13, 21, 42, 87, 100\}$; the rationale-trained ensemble contains the validation-selected seeds $\{100, 87, 42\}$. The equal-count direct-only control adds independently trained seeds $\{314, 2718, 2026\}$ and includes all eight members, without test-time pruning. The representative direct-baseline runs use LoRA rank 16, LoRA scaling 32, dropout 0.05, learning rate 10^{-4} , AdamW weight decay 0.01, warm-up ratio 0.05, three epochs, per-device batch size 1, and gradient accumulation 16. The headline rationale-trained runs use the same LoRA rank/scaling/dropout but batch size 2, accumulation 4, and learning rate 2×10^{-4} . The fresh headline-recipe direct-only control uses the latter recipe with the same three seeds and disables only the rationale objective. Exact run configurations are retained in JSON summaries.

We report accuracy, class F1, Macro-F1, expected calibration error (ECE) when available, paired percentile bootstrap intervals, and exact McNemar tests. Bootstrap samples are drawn at the claim level with 10,000 iterations. We call $P(\Delta > 0)$ a bootstrap probability of superiority, not a frequentist p -value. All test table deltas compare predictions with matched claim IDs.

5.3 Baselines and comparison discipline

Baselines include direct-only, rationale-trained-only, equal posterior averaging, symmetric confidence routing, no-asymmetric-NEI routing, disagreement-only routing, and max-confidence routing. Oracle-routing diagnostics are omitted because they do not constitute an implementable baseline. Literature scores are reported in a separate protocol-aware table. MOCHEG, HGTMFC, AMuFC, MEVER, M-RAV, and open-web systems do not necessarily share evidence, split, modality, or metric definitions; therefore a row is not treated as a direct leaderboard comparison unless its protocol is explicitly aligned.

5.4 Promotion gates and split discipline

We report a consistent directional-bootstrap reference threshold of .95, together with percentile intervals and exact McNemar tests; it is a reporting criterion rather than a claim of preregistration in this anonymized manuscript. The later duplicate-family-safe development studies use train-only folds, fixed training epochs, compute-matched controls, and fresh confirmation folds. A screen-only gain therefore cannot become an official-test claim. CURE-Ensemble is selected on validation and applied once to test. The post-hoc $\tau = .60$ sensitivity result is reported separately and never used as a

frozen-test result. The fixed AND condition uses $K = 5$ and $\tau = .49$; a later K sweep is descriptive rather than a retrospective change to this condition.

Multiplicity and inferential scope. The paper reports an expanding set of related development comparisons, recipe, teacher, and ensemble-size controls, raw/strict test-track contrasts, and one external negative-transfer audit. We do *not* apply Bonferroni, Holm, or FDR correction because $P(\Delta > 0)$ is a directional paired-bootstrap estimate rather than a frequentist p -value, and these comparisons do not form a prospectively specified family of independent hypothesis tests. This does not eliminate the risk of selection across many experiments. Accordingly, no individual bootstrap probability—including values near or above .95—is presented as familywise statistical significance, and all such values are accompanied by percentile intervals and, where available, exact McNemar tests. The paper’s claims are limited to frozen-policy point estimates and directional replication across raw/strict tracks; causal and superiority claims remain explicitly unresolved.

6 Main Results

6.1 Mechanism-focused evidence

We report only the development evidence needed to assess the proposed mechanism in the main paper. Earlier interventions are retained as a compact appendix ledger. They establish four negative findings: evidence compression did not reliably replace document context; symmetric auxiliary objectives and PCGrad did not generalize; calibration and unconstrained routing alternatives were unstable; and counterfactual curricula failed matched-control confirmation. These failures motivate retaining a competent direct verifier and treating the rationale-trained branch as a constrained NEI signal rather than a universally superior classifier.

6.2 Frozen validation and ablations

Reviewer controls. We additionally compare AND with a direct-only self-deferral baseline. The latter outputs NEI when $p_D(\text{NEI}) \geq .23$; its threshold is selected on official validation and frozen before either test track is read. Table 4 shows that AND exceeds this simpler uncertainty policy by .500 MF1 points on raw P1 and .523 points on strict P1. The directional advantage is replicated on both tracks, but its paired bootstrap

Development question	Confirmed observation
Evidence compression	Atomic evidence loses discourse context; a rationale-distilled selector does not exceed retrieval Top-3.
Symmetric multi-task learning	B6 auxiliary transfer and PCGrad fail their fresh-fold or bootstrap promotion gates.
Calibration and routing	Logit bias, cross-fit calibration, and unconstrained routers produce small, non-robust gains.
Counterfactual supervision	A +.01654 Fold-0 screen gain becomes -.00104 versus matched control in confirmation.

Table 2: Four negative-result families motivating the constrained design. Detailed phase-level results are in Appendix A.

Variant	Acc.	Macro-F1	F1-S	F1-R	F1-NEI
A0 Direct-only	.700549	.692045	.720348	.769904	.585882
A1 Rationale-trained-only	.706731	.701191	.704519	.780834	.618221
A2 Equal posterior average	.699863	.692773	.707071	.772608	.598639
A3 Symmetric confidence switch	.698489	.691095	.706145	.771930	.595211
A4 Unconstrained confident switch	.710165	.704436	.707904	.781915	.623491
CURE (AND)	.714973	.711205	.703016	.793008	.637591

Table 3: Frozen validation component ablations (validation is used only to evaluate fixed policies). “Unconstrained confident switch” selects the rationale-trained output of any class when its maximum probability is at least τ . CURE applies the class-constrained AND rule at $\tau = .49$. Degenerate disagreement-only and max-confidence variants are omitted from the main table and reported in the accompanying ablation artifact. F1-S/R denote Supported/Refuted.

probabilities (.7805/.7920) and intervals are not sufficient for a conventional significance claim. Thus, this control rules out a direct NEI threshold as a complete account of the observed AND point estimate, while leaving the magnitude of the incremental benefit uncertain.

Broader prior-work inventory. Table 5 lists twelve system rows: eight prior-work rows and four rows from this work. It is deliberately *not* a single leaderboard: a reported F-score is preserved as F-score rather than silently renamed Macro-F1; filtered tests, gold evidence, or alternate preprocessing are marked. Only rows labelled “aligned raw P1” support an apples-to-apples point-estimate comparison.

AND has the highest frozen-validation point estimate in Table 3. Relative to direct-only it gains 1.916 Macro-F1 points and 1.442 accuracy points. Replacing the class-constrained rule by an unconstrained confident switch reduces Macro-F1 by 0.677 points, whereas equal averaging loses 1.843 points. The rationale-trained-only branch is useful (+0.915 over A0) but remains 1.001 points be-

Track	Comparison	Base MF1	AND MF1	Δ	$P(\Delta > 0)$
Validation	AND vs. rationale-trained-only	.70119	.71121	+.01001	.9470
Validation	AND vs. unconstrained switch	.70444	.71121	+.00677	.8952
Raw P1	AND vs. direct	.54493	.55470	+.00977	.9529
Raw P1	AND vs. self-deferral	.54970	.55470	+.00500	.7805
Strict P1	AND vs. direct	.54490	.55433	+.00943	.9477
Strict P1	AND vs. self-deferral	.54911	.55433	+.00523	.7920

Table 4: Frozen-policy reviewer controls. Self-deferral uses only the direct expert with validation-selected $\tau_D = .23$; AND uses $K = 5, \tau = .49$. Raw/strict rows use no test labels for policy selection. The bootstrap probability is directional evidence, not a substitute for a confidence interval.

low AND. AND’s validation comparisons against rationale-trained-only ($P = .9470$) and unconstrained switching ($P = .8952$) do not independently clear the reported .95 directional-bootstrap reference threshold; they are point-estimate evidence, not decisive mechanism confirmation. The raw and strict P1 results retain the direction of the direct comparison, while their uncertainty intervals require a cautious interpretation.

6.3 Official test results and leakage control

6.3.1 Configuration provenance and frozen test results

Table 8 is the single configuration map for all raw-P1 scores that might otherwise be read as interchangeable direct baselines. It shows that the four direct-family values differ in seed count, recipe, inference artifact, or post-processing; the headline recipe is not a neutral replacement for the legacy direct recipe. On validation, its direct-only control is .00492 MF1 below the base-recipe direct A0 (.68712 vs. .69205); on raw test, its three-seed output is .01057 below the legacy five-seed value. This pattern is consistent with optimization sensitivity or altered class behavior under the higher-learning-rate recipe, and is an alternative explanation for some rationale-branch behavior. The matched-recipe comparison therefore isolates λ_r *within* that recipe but cannot establish that the headline recipe itself is harmless. Likewise, the 7B three-seed average (.53986) is lower than its best individual member (seed 100, .55128), so it should not be interpreted as a monotonic ensemble improvement.

The frozen CURE-Ensemble reaches Macro-F1 0.55507 and accuracy 0.57535 on raw P1 official test, compared with 0.54531 and 0.56798 for the previously materialized direct five-seed ensemble. The unadjusted paired bootstrap probability of improvement is 0.9839 with CI

[+0.00097, +0.01850]; following the multiplicity policy above, this is directional evidence rather than a familywise significance claim. Its score is higher than AMuFC’s reported raw-P1 point estimate (Macro-F1 .5400; accuracy .5460), but this is not a paired comparison and the two systems differ in retrieval, backbone, modality use, and member count. We therefore report it as a protocol-aligned point estimate rather than a claim of statistical superiority or state of the art. CURE-Student disagreement distillation reaches Macro-F1 0.54940 for a post-hoc reported single seed (87); it is an exploratory efficiency diagnostic, not a multi-seed or locked-test effectiveness comparison.

Table 10 closes the member-count confound with a new, internally paired canonical control. Adding three predeclared direct-only members improves the five-member canonical direct anchor by +.00444/.00467 MF1 on raw/strict P1 (95% CIs [−.00199, .01080]/[−.00183, .01126]; $P(\Delta > 0) = .9134/.9216$). At the same eight-member budget, the heterogeneous five-direct-plus-three-rationale ensemble has a further +.00647/+ .00503 MF1 point-estimate advantage (CIs [−.00257, .01556]/[−.00401, .01403]; $P(\Delta > 0) = .9187/.8611$). Thus the control establishes neither a statistically resolved heterogeneity advantage nor a null member-count effect. It does show that the heterogeneous ensemble’s reported point estimate cannot be attributed solely to adding members, while preserving the distinction between this canonical paired artifact and the earlier legacy CURE-Ensemble report.

AND has canonical test artifacts on both tracks. On raw P1, AND obtains .55470 Macro-F1 and .57125 accuracy, compared with its independently inferred direct anchor’s .54493 and .56798: $\Delta\text{MF1} = +.00977$, with 86 helpful and 78 harmful changes and paired bootstrap $P(\Delta > 0) = .9529$. On strict P1, it obtains .55433 versus .54490 ($\Delta\text{MF1} = +.00943$; 84/77 helpful/harmful; $P(\Delta > 0) = .9477$). The direct-only self-deferral control is weaker on both tracks (.54970 raw and .54911 strict); AND improves on it by +.00500/+ .00523 MF1, respectively. The associated probabilities (.7805/.7920) and all test-track percentile intervals include zero, and McNemar tests are non-significant. We therefore treat the replicated direction as modest directional test evidence, not as statistically resolved confirmation.

System	Venue/year	Evidence and evaluation	Passes	Acc.	Reported score	Comparison status
<i>Prior work: MOCHEG reports (raw-P1-aligned rows are explicitly marked)</i>						
MOCHEG (Yao et al., 2023)	SIGIR 2023	system text+image	–	–	F-score .4406	original benchmark; score label retained
LVL4M4FV (Tahmasebi et al., 2024)	arXiv 2024	retrieved text+image	–	.4510	micro-F1 .4510	macro not source-reported
HGTMFC (Pang et al., 2025)	AAAI 2025	retrieved text+image, raw test	–	.4861	Macro-F1 .4678	aligned split; different model/retrieval
MetaSum Perceiver (Chen et al., 2024b)	ACL 2024	system text+image	–	–	F-score .4860	source does not label it Macro-F1
MEVER (Zhang et al., 2026)	arXiv 2026	retrieved multimodal	–	.483±.021	MF1 .497±.012	custom consistent preprocessing
M-RAV (Qwen2.5-32B) (Luu et al., 2027)	IPM, 2027 issue	system evidence, filtered $n = 2001$	–	.5002	MF1 .5014	filtered test; not aligned
Entailed Opinion (Kumar et al., 2025)	arXiv 2025	domain-generalization setting	–	–	MF1 .5700	not P1 system retrieval
AMuFC (Jung et al., 2026)	HCAI-CIKM 2025; arXiv 2026	retrieved text+image, raw test	–	.5460	MF1 .5400	raw-P1 aligned challenger
<i>This work: frozen raw official P1 ($n = 2,442$)</i>						
CURE Direct ensemble	this work	fixed corpus; direct argmax	5	.5680	MF1 .5453	frozen reference artifact
CURE rationale seed 100	this work	fixed corpus; rationale-trained	1	.5696	MF1 .5513	single-member diagnostic
CURE AND	this work	fixed corpus; frozen $K = 5, \tau = .49$	8	.5713	MF1 .5547	canonical constrained router
CURE-Ensemble	this work	fixed corpus; validation-selected blend	8	.5754	MF1 .5551	frozen-test variant

Table 5: Protocol-aware comparison with prior work and frozen CURE variants. “Passes” is the number of sequential member forward passes when reported; “–” means not source-reported. Raw-P1 alignment means the raw MOCHEG test with system-retrieved evidence, not identical retrievers, modalities, backbones, or capacity. Thus, aligned rows support point-estimate context only, not a significance or state-of-the-art claim.

Evidence policy	Acc.	Rationale MF1	AND MF1
$K = 1$	.707418	.660969	.704452
$K = 3$	.718407	.695423	.715270
$K = 5$	.714973	.701191	.711205
Direct-only	.700549	–	.692045

Table 6: Evidence-count sensitivity on validation. The fixed AND condition uses $K = 5$, inherited from the direct retrieval protocol. The later $K = 3$ sweep has the highest validation score but is not substituted for the fixed condition or used for an official-test claim.

τ	Validation MF1	Status
.40	.70774	sensitivity
.45	.70932	sensitivity
.49 (frozen)	.71121	registered AND
.50	.71109	sensitivity
.55	.71089	sensitivity
.60	.71012	sensitivity only
.65	.70755	sensitivity
.70	.70420	sensitivity

Table 7: Validation threshold sensitivity for AND. $\tau = .49$ is frozen for the registered condition. No raw-test threshold sweep is used for parameter selection.

6.3.2 Artifact alignment

The .00039 raw-MF1 difference between the legacy direct ensemble (.54531) and the independently inferred AND anchor (.54493) arises because the latter is a new canonical raw-manifest inference artifact built to align exactly with the AND members.

It is not used in a cross-artifact paired comparison: every AND comparison uses only predictions materialized together on the same manifest. Because raw and strict ranked evidence differ for some shared IDs, each track is inferred independently from the same frozen checkpoints; no intersection

Raw-P1 output	Acc./MF1	Seeds	Training configuration	Decision output	Role; comparability constraint
Legacy direct ensemble	.56798/.54531	13,21,42,87,100	Base direct recipe: $b = 1$, $\text{accum.}=16$, $\text{lr}=10^{-4}$	direct argmax	Legacy frozen benchmark; used for CURE-Ensemble reporting
Canonical direct anchor	.56798/.54493	13,21,42,87,100	Same checkpoints and recipe as legacy direct	direct argmax	Independently materialized to align exactly with AND; only paired comparator for AND
Canonical direct-only size control	.57207/.54936	legacy 5 + 314,2718,2026	Base direct recipe for all eight members	average probabilities then argmax	Equal-count B22 control; internally paired with canonical heterogeneous 5+3 only
Canonical heterogeneous 5+3	.57617/.55584	5 direct + rationale 42,87,100	Five base-recipe direct plus three headline rationale members	average probabilities then argmax	Equal-count B22 comparator; a separately materialized control artifact, not a replacement for legacy CURE-Ensemble reporting
Direct self-deferral	-.54970	13,21,42,87,100	Same base direct checkpoints	direct, except NEI if $p_D(\text{NEI}) \geq .23$	Frozen simpler routing control; not a new trained direct model
Headline direct-only	.55569/.53474	42,87,100	Headline recipe: $b = 2$, $\text{accum.}=4$, $\text{lr}=2 \times 10^{-4}$, $\lambda_r = 0$	average probabilities then argmax	Matched-recipe causal control; not comparable to five-seed legacy score
Rationale seed 100	.56962/.55128	100	Headline recipe with 7B teacher rationales, $\lambda_r = .25$	direct argmax	Single-member diagnostic; not an ensemble comparison
Rationale 7B ensemble	.55979/.53986	42,87,100	Same headline rationale recipe, 7B teacher, $\lambda_r = .25$	average probabilities then argmax	Matched-recipe and capacity-control reference; averaging is worse than seed 100
Rationale 3B ensemble	.56839/.54829	42,87,100	Same headline rationale recipe, 3B teacher, $\lambda_r = .25$	average probabilities then argmax	Matched teacher-capacity control; no routing policy fitted on test
Rationale Mistral-7B ensemble	.56061/.53965	42,87,100	Same headline rationale recipe, Mistral-7B teacher, $\lambda_r = .25$	average probabilities then argmax	Matched teacher-family control; no routing policy fitted on test

Table 8: Configuration map for raw P1 outputs that could otherwise be called “direct” or “rationale-trained.” Scores differ because seed sets, optimization recipes, ensemble operators, and/or post-processing differ. Only rows explicitly designated as a paired comparison share a common prediction artifact and should be compared by a paired statistic.

or test-time tuning is used.

6.3.3 Validation–test gap

The larger validation–test difference is not interpreted as a model improvement. Validation was used for seed and threshold development and has a distinct claim composition. The contraction from $+0.01916$ MF1 on validation to $+0.00977/+0.00943$ on raw/strict test is consistent with development-selection sensitivity, but this study cannot apportion it among distribution shift, optimization, and selection.

6.3.4 Routing versus averaging

Raw CURE-Ensemble (.55507) is marginally above raw AND (.55470), although A2 averaging is unhelpful on validation. This difference does not establish that either decision rule dominates: the systems have distinct validation-frozen compositions and objectives, and neither is an ablation of the other. The later equal-count direct-only control similarly retains a heterogeneous 5+3 point-estimate advantage, but its intervals cross zero. Both reports record manifest, retrieval, input-prediction, and materialized-output hashes.

7 Additional Ablations and Negative Results

7.1 Sentence-selection branch

A sentence-selection ablation distills teacher key-passage attribution into a MiniLM cross-encoder that adaptively selects one to three sentences. It is an evidence-selection ablation, not the dual-expert router. On Fold 0, rationale-distilled selection improves a generic cross-encoder by $+0.00865$ MF1 but remains -0.00214 behind the retrieval Top-3 control. It was therefore not expanded to more seeds or test evaluation (Table 11). This result prevents us from attributing the rationale-trained-expert effect to sentence compression alone; teacher capacity is instead isolated by the matched full-train ablation below.

7.2 Why earlier constraint approaches were not promoted

Earlier interventions supplied useful diagnostics but did not establish a stable main method. A source-aware auxiliary ensemble reached .7080 MF1 on validation, yet its bootstrap probability

Family	Frozen system	Passes	Accuracy	Macro-F1	Δ MF1	Evaluation role and comparison constraint
Direct	Direct five-seed ensemble	5	.5680	.5453	–	Legacy frozen baseline artifact.
Rationale-trained	Seed 42	1	.5524	.5338	–	Individual member; not a multi-seed comparison.
	Seed 87	1	.5606	.5394	–	Individual member; not a multi-seed comparison.
	Seed 100	1	.5696	.5513	–	Best single rationale-trained member.
Probability ensemble	Full 10-member	10	.5717	.5512	+0.0059	Unpruned control; fixed averaging.
	Selected 8-member	8	.5754	.5551	+0.0098	Validation-selected, frozen-test benchmark variant.
Constrained routing	CURE AND (5+3 experts)	8	.5713	.5547	+0.0098 [†]	Canonical frozen router; [†] denotes comparison to jointly materialized direct anchor (.54493), not legacy artifact.
Compression	CURE-Student DKD seed 87	1	.5676	.5494	–	Post-hoc single-seed efficiency diagnostic.

Table 9: Raw official P1 test ($n = 2,442$): frozen CURE system variants. Bold marks the highest point estimate in each reported metric, not a claim of statistical superiority. CURE-Ensemble and AND have distinct validation-frozen compositions and objectives; they are not interchangeable ablations. See Table 8 for the complete configuration provenance.

Track	Ensemble	Accuracy	MF1	F1-S	F1-R	F1-NEI
Raw P1	Canonical direct (5)	.56798	.54493	.58340	.65106	.40032
Raw P1	Direct-only size control (8)	.57207	.54936	.59011	.65225	.40573
Raw P1	Heterogeneous CURE (5+3)	.57617	.55584	.57969	.65794	.42988
Strict P1	Canonical direct (5)	.56861	.54490	.58321	.65278	.39872
Strict P1	Direct-only size control (8)	.57272	.54958	.58994	.65366	.40512
Strict P1	Heterogeneous CURE (5+3)	.57560	.55461	.57948	.65852	.42583

Table 10: Predeclared equal-member-count control. The direct-only size control contains the five canonical direct members plus direct seeds 314, 2718, and 2026; all eight receive equal probability weight. Heterogeneous CURE uses the same five direct members and three frozen rationale-trained members (42, 87, 100), also at equal weights. Direct-8 minus direct-5 is +.00444/+ .00467 MF1 (95% CIs $[-.00199, .01080]/[-.00183, .01126]$; $P(\Delta > 0) = .9134/.9216$); heterogeneous 5+3 minus direct-8 is +.00647/+ .00503 (CIs $[-.00257, .01556]/[-.00401, .01403]$; $P(\Delta > 0) = .9187/.8611$), raw/strict respectively. No test-time member pruning or weighting is used.

Evidence policy	Acc.	MF1	F1-S	F1-NEI
Original Top-5 anchor	.6664	.6435	.6026	.5067
Retrieval Top-3	.6720	.6482	.5964	.5187
Generic CE selector	.6496	.6374	.5674	.5253
Rationale-distilled selector	.6651	.6460	.6098	.5103

Table 11: Sentence-selection train-only Fold-0 screen ($n = 2,326$). The rationale-distilled selector is better than generic CE but does not exceed retrieval Top-3.

against the compute-matched direct control was .9354, below the reported .95 reference threshold. Calibration and curriculum variants exhibited strong Fold-0 gains but degraded under fresh-fold confirmation. A counterfactual verdict curriculum improved Fold 0 by +.01654 MF1 versus its anchor, but aggregated confirmation was -.00104 versus the matched control. These negative

outcomes motivate the CURE design choice: preserve a competent direct decision model and use a rationale-trained complementary signal through a constrained interface rather than continuously modifying the anchor’s decision boundary.

7.3 Matched-recipe explanation supervision control

This is a distinct control from the headline-recipe contrast in Table 12. We ran a five-fold train-only OOF experiment in which verdict-only and explanation-supervised models use one common training script and optimizer schedule. The explanation model obtains .65757 aggregate MF1 versus .65762 for the matched verdict-only control ($\Delta = -.00006$; bootstrap $P(\Delta > 0) = .4940$). Applying the frozen asymmetric router to this

Ensemble	Acc.	MF1	F1-S	F1-R	F1-NEI
Headline-recipe direct-only	.69437	.68712	.69874	.77003	.59259
Rationale-trained	.70673	.70119	.70452	.78083	.61822

Table 12: Headline-recipe direct-only control on official validation ($n = 1,456$). Rationale-trained minus direct-only: $\Delta\text{MF1} = +.01407$, helpful/harmful=88/70, bootstrap 95% CI $[-.00369, +.03180]$, and $P(\Delta > 0) = .9396$. Both arms use validation for checkpoint selection; no test result is selected or evaluated here.

matched explanation expert yields .65898, only +.00135 over the control ($P(\Delta > 0) = .7273$). These OOF values are neither validation nor test results and cannot be directly compared with the .71121 validation policy score. They show that structured rationale loss alone does *not* improve average verdict accuracy under that common recipe.

7.4 Headline-recipe direct-only control

To reduce the optimization-recipe confound, we train a new verdict-only ensemble with the exact headline rationale-trained recipe: the same data, no gold injection, batch size 2, gradient accumulation 4, learning rate 2×10^{-4} , three epochs, top- $K = 5$, and seeds $\{42, 87, 100\}$. The only intended difference is $\lambda_r = 0$ versus .25. On validation, the rationale-trained ensemble is higher by +.01407 MF1 and +.01236 accuracy, with its largest class gain on NEI (+.02563; Table 12). This directional contrast reduces the explanation-versus-recipe ambiguity, but does not settle it statistically: the paired 95% interval includes zero, $P(\Delta > 0) = .9396$, and both arms selected their best checkpoint on the same official validation split.

We then freeze these same three seeds and independently infer raw and strict P1 predictions. The rationale-trained ensemble retains a positive point-estimate difference on raw (+.00512 MF1) and strict (+.00685 MF1) test (Table 13); the directions therefore replicate across both test manifests. However, their intervals contain zero ($P(\Delta > 0) = .7694/.8270$). The full-train/test control is consequently evidence compatible with a rationale contribution, rather than statistically resolved causal confirmation.

7.5 Matched teacher capacity and family ablations

Teacher alternatives are tested directly rather than inferred from the headline result. We regenerate only the training rationales, retrain the same Qwen3-4B students with the same expla-

Track	Direct MF1	Rationale MF1	Δ	Helpful/Harmful	95% CI	$P(\Delta > 0)$
Raw P1	.53474	.53986	+00512	119/109	$[-.00856, .01876]$	.7694
Strict P1	.53222	.53906	+00685	120/106	$[-.00705, .02049]$	.8270

Table 13: Frozen full-train headline-recipe confirmation. The same three validation-frozen seeds are independently inferred on raw and strict P1; no test label selects a checkpoint, seed, or policy. Positive test point estimates replicate, but neither paired interval excludes zero.

nation weight, data, retrieval, prompts, seeds $\{42, 87, 100\}$, optimizer recipe, and validation checkpoint protocol, and independently infer both P1 tracks. Thus, each ablation changes teacher capacity or family but does not add data, gold evidence, student capacity, or test-time tuning.

The detailed results are in Appendix G. Within Qwen2.5, the 3B teacher has higher raw/strict point estimates by .00844/.00753 MF1 relative to 7B, but the paired 95% intervals contain zero and $P(7B > 3B) = .1166/.1365$. Across teacher families at approximately 7B parameters, Mistral-7B differs from Qwen2.5-7B by only $-.00021/- .00099$ MF1 on raw/strict P1 ($P(\text{Mistral} > \text{Qwen}) = .4817/.4454$; both intervals contain zero). These matched comparisons neither support a simple capacity or tested-family account of the rationale branch’s behavior, nor establish invariance beyond the two families and capacities evaluated.

8 Efficiency and Robustness Analysis

8.1 Inference efficiency

On an RTX 5090 with 256 common validation claims, batch size 4, and two warm-up batches, direct, rationale-trained, AND, and the one-model student require 320.72, 193.17, 513.88, and 64.47 ms per claim, respectively (Table 14). AND is therefore $1.60\times$ slower than its five-pass direct anchor for only a +.00500/+ .00523 unresolved advantage over direct self-deferral on raw/strict test. The student uses one pass instead of the router’s eight sequential passes. All ensemble timings are sequential; the repeated 14.74 GiB figure is the maximum allocated memory of one loaded member, not a comparison of total ensemble memory. Figure 3 provides descriptive quality–cost context: its raw-P1 effectiveness values and validation-subset timing values are deliberately not treated as a paired effectiveness–latency comparison. Subset effectiveness scores produced by this benchmark are diagnostic only.

System	Passes	ms/claim	claims/s	Peak GiB
Direct ensemble	5	320.72	3.118	14.74
Rationale-trained ensemble	3	193.17	5.177	14.74
AND dual ensemble	8	513.88	1.946	14.74
CURE-Student	1	64.47	15.511	14.74

Table 14: Efficiency benchmark: RTX 5090, 256 claims, batch size 4, two warm-up batches. Model-loading time is excluded.

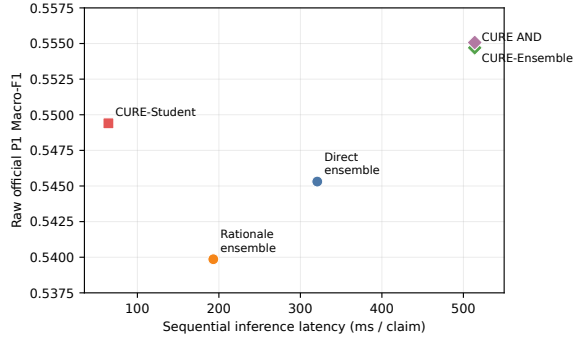


Figure 3: Descriptive compute-quality context. Macro-F1 is the reported raw-official P1 point estimate; latency is sequential inference measured on a common 256-claim validation subset. The axes therefore summarize two frozen artifacts and do not constitute a paired effectiveness-cost test.

8.2 Subgroup behavior

On the frozen validation audit, AND improves Macro-F1 by 1.916 points, with 58 helpful and 37 harmful changes. Table 15 separates where this gain occurs. It is largest for NEI (+3.239), rank-one evidence (+2.772), evidence hits (+2.280), and Snopes (+1.994). Retrieval misses decrease by 3.223 points and Supported claims decrease by 1.404 points. These are post-hoc descriptive groups, based partly on diagnostics unavailable at inference. They do not establish that the rationale-trained posterior measures corpus-level insufficiency, particularly under retrieval failure.

The direct and rationale-trained experts agree on 85.44% of validation verdicts (Cohen’s $\kappa = .7756$). Their errors are also strongly associated ($\kappa = .7184$; Yule’s $Q = .9561$), so “complementary” does not mean independent. Still, 90 claims are correct only for the rationale-trained expert and 81 only for the direct expert (Appendix D). This limited asymmetric overlap is consistent with a constrained deferral rule, but is not evidence that arbitrary expert switching will be safe.

Group	n	Δ MF1	Helpful/Harmful
Overall	1,456	+0.01916	58 / 37
PolitiFact	431	-.00249	0 / 6
Snopes	1,025	+0.01994	58 / 31
Gold NEI	478	+0.03239	58 / 0
Gold Refuted	487	-.00343	0 / 9
Gold Supported	491	-.01404	0 / 28
Gold rank 1	1,233	+0.02772	54 / 25
Gold rank 2–5	131	-.02607	3 / 7
Gold not retrieved	59	-.03260	1 / 4

Table 15: Selected frozen-validation subgroup audit: AND minus direct. Gold rank is a diagnostic label unavailable at inference, not a routing input.

Fold-0 variant	Acc.	MF1	F1-S	F1-NEI
Matched control	.6690	.6570	.6013	.5368
Naive ensemble KD	.6707	.6513	.6106	.5152
Counterfactual only	.6758	.6577	.6063	.5332
Full KD + disagree + CF	.6776	.6584	.6022	.5359
Disagreement KD	.6836	.6687	.6276	.5472

Table 16: CURE-Student train-only Fold-0 mechanism screen ($n = 2,326$). The five-fold confirmation uses disagreement KD only; counterfactual terms are reported solely as screen ablations.

8.3 CURE-Student distillation

Five-fold train-only OOF confirmation gives the student control Macro-F1 0.660788 and disagreement-KD Macro-F1 0.667607, a +0.006819 difference. Four of five folds improve and paired bootstrap $P(\Delta > 0) = 0.9517$. Table 16 shows why the disagreement target matters: naive ensemble KD is worse than the matched control, while counterfactual loss alone is nearly neutral and the full objective reduces the screen gain. This supports CURE-Student as a practical compression extension, but its margin is close to the reported bootstrap reference threshold. The reported seed-87 test number is post-hoc descriptive among the evaluated student seeds, not a locked-test model-selection claim.

9 Error Analysis

The dominant helpful transition is direct Refuted/Supported to rationale-trained NEI on gold NEI claims. Harmful transitions are direct Supported to NEI (28 cases) and direct Refuted to NEI (9 cases). This explains why AND improves NEI while reducing decisive-class F1. Table 24 gives the aggregate audit and Table 25 gives deterministically selected raw-test examples, including non-repairing both-wrong transitions. The failure atlas also shows that routing under missing or rank-2–5 evidence is less reliable. A future system should estimate retrieval reliability before allowing deferral,

but using gold rank or QREL status at inference would be invalid.

10 Discussion

The results support a narrow interpretation of the contribution. Rationale-trained and direct experts form a complementary pair, but explanation supervision alone is not consistently beneficial: a matched explanation-loss ablation produced approximately zero aggregate gain. The headline-recipe control is directionally favorable to rationale training, yet its validation and test intervals include zero. Crucially, this limited contrast is recipe-conditional: $\lambda_r > 0$ may transfer structured rationale information, or it may partly regularize adverse behavior induced by the headline optimization schedule. It therefore does not establish a recipe-invariant value of rationale supervision. The matched teacher controls further weaken simple teacher explanations: replacing Qwen2.5-7B with Qwen2.5-3B raises raw/strict point estimates by .844/.753 MF1, while replacing it with same-scale Mistral-7B changes them by only $-.021 / -.099$ MF1. Neither set of paired intervals resolves statistically. Accordingly, the evidence does not identify a causal mechanism within rationale supervision, and it does not support the claim that a larger or one tested teacher family is responsible for the branch’s behavior.

Moreover, a decisive-evidence coverage diagnostic finds that the direct posterior has stronger ranking and calibration than the rationale-trained posterior (AUROC .6486 vs. .6155; Brier .1122 vs. .1406); this diagnostic uses gold labels only after prediction and is never a router input. The new equal-count ensemble control makes the interpretation more specific: member count itself has a positive but unresolved direct-8 versus direct-5 effect, while the heterogeneous 5+3 ensemble remains directionally higher at the same count. AND also exceeds a frozen direct self-deferral baseline on both test tracks, although that incremental comparison is not statistically resolved. The useful object is therefore the asymmetric interface between complementary class posteriors, not a demonstrated sufficiency estimator. This interface also exposes a compute–quality trade-off: AND uses eight sequential expert passes, whereas CURE-Student compresses complementary teacher behavior into one student. A stronger external definition of evidence sufficiency remains necessary before claiming that

either posterior is calibrated for that construct.

Frozen-policy external transfer. We additionally transfer the frozen MOCHEG policy ($K = 5, \tau = .49$) without retraining or threshold re-tuning to a 300-claim SciFact development audit (Wadden et al., 2020). Retrieval is a deterministic corpus-wide scikit-learn TF-IDF ranker (lower-cased, Unicode-accent stripped, word 1–2 grams, sublinear TF, L_2 normalization); it retains the top five abstracts directly, with no reranker. Cited documents, gold rationales, and external labels are unavailable to either expert at inference. This is intentionally not a SciFact leaderboard comparison: its standard task requires document- and sentence-level rationale outputs, while the public development split is used here solely for retrospective evaluation. The transfer fails decisively (Appendix E): AND is $-.08310$ MF1 below the direct expert (95% CI $[-.13260, -.03444]$; exact McNemar $p = .0113$; $P(\Delta > 0) = .0001$). The rule activates on 132/300 claims (44.0%), but only 45 activations alter the direct output; these changes repair 10 and harm 26. The rationale-trained branch’s Refuted F1 falls to .2133, compared with .4694 for the direct ensemble, so broad deferral transfers this domain-specific failure. This negative result bounds our claim: the frozen MOCHEG policy does not generalize zero-shot across this domain and retrieval protocol.

11 Limitations and Ethical Considerations

MOCHEG is a single benchmark with two principal sources and potentially duplicated claims; we therefore report raw and strict tracks. The fixed-corpus protocol does not measure live-web robustness or establish multimodal effectiveness. The rationale-trained expert inherits teacher errors and pseudo-label noise; moreover, teacher rationales are conditioned on the gold *training* verdict, which is appropriate for training but prevents the rationale channel from being interpreted as independent supervision. The matched teacher controls span two Qwen2.5 capacities and one Mistral family, but all results have intervals containing zero and cannot identify why target distributions differ or establish broader teacher-family invariance. The headline-recipe direct control reduces causal ambiguity and preserves a positive point estimate on both test tracks, but its validation and test intervals include zero and its inference is conditional

on that optimization recipe. The new equal-count direct-only control removes the previously missing eight-member baseline, yet both its member-count and heterogeneous-versus-direct-8 intervals include zero. The multiple related development and test comparisons create a real selection/multiplicity risk; we therefore make no familywise significance claim and report all inspected alternatives, including negative results. Subgroup analyses are descriptive and do not justify post-hoc policy tuning. The paired binary-correctness power approximation in Appendix D gives 80% minimum detectable accuracy effects of .01469/.01460 on raw/strict P1, substantially larger than the observed .00328/.00288 effects; it is not a Macro-F1 power calculation, but makes the limited resolution of the current test sets explicit. Although AND now has independently inferred raw and strict test artifacts, its 95% paired-bootstrap intervals include zero and strict-track directional evidence is .9477; stronger multi-dataset confirmation is needed before broad generalization claims. Our only external audit finds statistically resolved negative zero-shot transfer to SciFact under TF-IDF retrieval; it is not a leaderboard comparison and does not establish that adaptation would fail, but it rules out a claim that the frozen MOCHEG policy is domain-general. Properly adapted multi-dataset experiments and more diverse teacher families remain future work. Automated fact checking can amplify source and retrieval bias; predictions should be treated as decision support, not authoritative adjudication.

12 Conclusion

We presented CURE, a dual-expert constrained routing framework for retrieved-evidence fact verification. Its AND rule improves frozen-validation Macro-F1 by 1.92 points and has raw/strict P1 test point-estimate gains of .977/.943 points under a fixed $K = 5$, $\tau = .49$ policy. A same-recipe direct-only control retains positive raw/strict rationale-trained differences of .512/.685 MF1 points, but their intervals include zero. Matched 3B-versus-7B and Mistral-7B-versus-Qwen2.5-7B teacher controls do not support simple capacity or tested-family explanations: their point estimates vary, but all paired intervals remain unresolved. The new eight-member direct-only control shows a positive member-count effect but leaves a further +.647/+ .503 MF1 heterogeneous-ensemble point-estimate advantage under an equal inference budget; those

intervals also include zero. A frozen zero-shot SciFact transfer audit is statistically resolved in the adverse direction, limiting the contribution to MOCHEG P1 rather than demonstrating domain-general routing. Thus the routing, recipe-control, teacher-control, and size-control evidence should be read as directional rather than statistically resolved causal proof. CURE-Ensemble provides a separate validation-selected benchmark variant, while CURE-Student demonstrates a one-model compression extension. The next research steps are adapted multi-dataset evaluation, broader teacher-family controls, and sharper retrieval-reliability gating, rather than post-hoc policy tuning on P1.

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A Detailed Development Ledger

Table 17 expands the main trajectory with the numerical evidence used to close or advance each hypothesis. These are reported for reproducibility and to prevent failed development screens from being misrepresented as official-test improvements.

B Five-Fold CURE-Student Confirmation

C Raw versus Strict Test Reporting

The raw official test ($n = 2,442$) is the primary literature-comparison track. The strict test ($n = 2,434$) removes eight verbatim cross-split duplicates and is a robustness track only. Table 19 does not merge the two populations. Canonical AND uses independently inferred predictions on each retrieval manifest because their ranked evidence IDs differ for some shared claims; every prediction file is checked against its full manifest rather than evaluated on an intersection.

D Complementarity and Diagnostic Resolution

These frozen-validation diagnostics describe the expert relationship and are not used to alter the router. The rank cross-tab uses gold evidence IDs only after prediction, so it cannot be interpreted as an available inference feature.

E Frozen External-Transfer Audit

We ran one deliberately strict out-of-domain audit, not a second benchmark comparison. The frozen MOCHeg direct ensemble (five members), rationale-trained ensemble (three members), top- $K = 5$, and AND threshold $\tau = .49$ were applied without SciFact training, seed selection, threshold retuning, cited-document access, or gold-rationale use. We use the 300 labeled claims in the public SciFact development split only to evaluate the already-frozen outputs (Wadden et al., 2020). Retrieval uses scikit-learn TF-IDF on corpus abstracts (lowercase, Unicode accent stripping, word 1–2 grams, sublinear TF, L_2 normalization), retaining its top five without reranking. Hence the results in Table 23 are not comparable to SciFact’s standard abstract-/sentence-rationale leaderboard, which uses a different output contract and typically trained in-domain systems.

The error profile provides a concrete boundary condition. The rationale-trained ensemble’s Re-

futed F1 drops by .2561 relative to direct, and AND activates on 44.0% of claims. Activation is especially misallocated to decisive gold classes: 44/64 Refuted claims (68.8%) and 35/124 Supported claims (28.2%) activate, compared with 53/112 NEI claims (47.3%). Of the 45 output changes, 13 correct Refuted and 13 correct Supported direct predictions are changed to NEI; only 10 NEI errors are repaired. Because 87 activations begin from a direct NEI prediction, they leave the output unchanged. We retain the failure rather than selecting a SciFact-specific alternative, because doing so would answer an adaptation question rather than the frozen-policy transfer question.

F Qualitative Routing Patterns

Table 24 reports a frozen-validation transition audit of the direct and AND predictions. It makes the aggregate trade-off explicit. Table 25 then gives four claim-level illustrations from the independently inferred raw official test artifact. They were selected as the lexicographically first IDs in four predeclared transition strata; gold labels are used only for retrospective reporting. Thus, the examples neither tune the router nor establish a separate effectiveness result.

G Teacher Capacity and Family Ablations

The capacity control replaces only Qwen2.5-7B-Instruct, which generates the structured training rationales, with Qwen2.5-3B-Instruct. The family control instead replaces it with Mistral-7B-Instruct-v0.3. Student backbone, student recipe, explanation weight, retrieved passages, prompts, seed set $\{42, 87, 100\}$, and validation checkpoint protocol are held fixed. The tests are independently inferred for their respective raw and strict retrieval manifests; no test label selects a teacher, checkpoint, seed, or policy.

H Reproducibility Checklist

The anonymized artifact package records manifests, retrieval signatures, prediction files, training summaries, seeds, and protocol flags. It includes training, component-ablation, subgroup, distillation, efficiency, and canonical-evaluation entry points. Before camera-ready release, we will add exact GPU driver versions, package-lock information, author metadata, and a public archival link.

ID	Intervention	Quantitative observation	Interpretation
SV	flat versus hierarchical sufficiency verification	Flat MF1=.6371; SV=.6398 (+.0027) on Fold 0; five-fold confirmation +.00060 $\pm$.00745	Hierarchy had small local benefit but no stable promotion.
DRO	source/evidence DRO	qrel-absent MF1 +.1004; overall MF1 -.0021	Robustness improvement cannot be bought with aggregate regression.
Atomic	atomic sentence verifier	sentence retrieval Recall@8=.8661; verdict MF1 -.0270 vs article evidence	Retrieval coverage alone is insufficient; discourse matters.
Packets	context packets	best interpolation +.01014 on one validation split; five-seed packet gain +.00387 $\pm$.00862	Complementarity exists but is not a reliable replacement.
Multi-task	sufficiency/polarity multi-task	auxiliary vs matched control +.00886 $\pm$.01081; bootstrap .9354	Positive transfer signal, below the reported reference threshold.
PCGrad	PCGrad	conflict rate about 45–46% in epoch 1; ensemble -.0105 vs control	Gradient conflict was real but projection did not improve generalization.
Router	selective expert router	MF1 +.00460; bootstrap .9294	Helpful precision, insufficient gain and confidence.
Bias	class-logit bias	MF1 +.00216; bootstrap .7604	Prior shift does not explain the main error pattern.
Ensemble	fixed seed ensemble	Fold-0 +.01348; fresh-fold mean +.00398	A screen peak is not confirmation.
Calibration	cross-fitted calibrator	aggregate MF1=.67054; +.00345 vs unweighted ensemble	Bootstrap/source gates fail.
Provenance	provenance calibrator	-.00176 vs unweighted ensemble	Provenance-conditioned calibration harms aggregate accuracy.
Constraints	joint constraints from base	+.00538 vs compute control but -.01376 vs anchor	Constraint tasks help relative to compute but induce anchor drift.
Curriculum	curriculum/blend	Fold-0 blend +.00510; five-fold mean +.00090	Direct curriculum is unstable.
NEI escape	NEI escape analysis	MF1 +.00209; accuracy +.01126	It helps supported/refuted escape but damages NEI under qrel absence.
Value gate	value-of-information gate	MF1 +.00380; AUROC=.5660	Observable utility is too weak for reliable routing.
Counterfactual	counterfactual omission	Fold-0 +.01654 vs anchor; confirmation -.00104 vs control	The strongest screen still fails matched-control confirmation.
Atlas	failure atlas	candidate/control .66517/.66621; help/harm 550/561	No stable subgroup justified further tuning on inspected folds.

Table 17: Earlier negative and diagnostic experiments. Each value remains in its native protocol; no row is a P1 official-test claim.

Fold	Control MF1	DKD MF1	Δ
0	.657035	.668656	+.011621
1	.659998	.656767	-.003231
2	.662265	.667143	+.004878
3	.647130	.660188	+.013058
4	.674931	.681645	+.006714
Aggregate	.660788	.667607	+.006819

Table 18: CURE-Student train-only out-of-fold confirmation. Four of five folds improve; the aggregate paired bootstrap probability is .9517.

System	Track	Acc.	MF1
Direct five-seed	raw official	.56798	.54531
Direct five-seed	strict	.56902	.54581
CURE-Ensemble	raw official	.57535	.55507
CURE-Ensemble	strict	.57477	.55383
Canonical direct-only eight	raw official	.57207	.54936
Canonical direct-only eight	strict	.57272	.54958
Canonical heterogeneous 5+3	raw official	.57617	.55584
Canonical heterogeneous 5+3	strict	.57560	.55461
CURE-Student seed 87	raw official	.56757	.54940
CURE-Student seed 87	strict	.56615	.54733
AND router	raw official canonical	.57125	.55470
AND router	strict canonical	.57149	.55433

Table 19: Track-separated test results. AND uses fixed $K = 5$, $\tau = .49$ and independently inferred raw/strict predictions; it is not an intersection evaluation. The canonical B22 rows are a separately materialized, internally paired size-control artifact and are not substituted for the legacy CURE-Ensemble report.

Direct \ rationale-trained	S	R	N
Supported	341	19	68
Refuted	13	585	58
NEI	18	36	318

Correctness overlap	Claims
Both correct	939
Direct-only correct	81
Rationale-only correct	90
Both wrong	346

Table 20: Frozen-validation expert complementarity. Rows of the left matrix are direct predictions and columns are rationale-trained predictions; S/R/N denote Supported/Refuted/NEI. Agreement is .8544, verdict Cohen’s $\kappa = .7756$, error Cohen’s $\kappa = .7184$, and Yule’s $Q = .9561$. The experts are strongly correlated but have a small asymmetric correctness overlap.

Gold-NEI retrieval status	n	Direct accuracy	AND accuracy	Δ	Routes / changes
QREL absent	7	.2857	.2857	+.0000	2 / 0
Gold not retrieved	11	.3636	.4545	+.0909	5 / 1
Gold rank after 5	2	.5000	.5000	+.0000	1 / 0
Gold rank 2–5	39	.6154	.6923	+.0769	25 / 3
Gold rank 1	419	.5203	.6492	+.1289	250 / 54

Table 21: Frozen-validation cross-tab for gold NEI claims. Gold-rank information is retrospective only and unavailable to AND. The largest observed NEI accuracy increase is in rank-one evidence, but this is a descriptive subgroup result rather than a routing rule.

Track	Observed Δ Acc.	80% MDE	Approx. n at effect
Raw official	+.00328	.01469	49,116
Strict	+.00288	.01460	62,771

Table 22: Paired-design power approximation for AND versus direct. It uses observed helpful/harmful correctness discordance and a two-sided $\alpha = .05$ normal approximation. This is not a power calculation for Macro-F1.

System	Acc.	MF1	Supp.	Ref.	NEI
Direct (5)	.5600	.5283	.6689	.4694	.4466
Rationale-trained (3)	.5067	.4330	.6403	.2133	.4453
Frozen AND	.5067	.4452	.6394	.2500	.4462

Table 23: SciFact zero-shot development audit ($n = 300$). All labels are used only after inference. AND is $-.08310$ MF1 versus direct (95% CI $[-.13260, -.03444]$; McNemar $p = .0113$; directional bootstrap probability .0001). It activates 132 deferrals, 45 of which alter the direct output, yielding 10 helpful and 26 harmful changes. It is an external negative-transfer result, not a SciFact leaderboard result.

Gold label	Direct prediction	AND prediction	Audit effect	Count	Reading of the transition
NEI	Refuted	NEI	helpful	33	Correctly defers a decisive direct error to NEI.
NEI	Supported	NEI	helpful	25	Correctly defers a second direct-error type to NEI.
Supported	Supported	NEI	harmful	28	Incorrect deferral of an already correct decisive prediction.
Supported	Refuted	NEI	both wrong	14	Changes one incorrect decisive label to another incorrect label.
Refuted	Refuted	NEI	harmful	9	Incorrect deferral of an already correct decisive prediction.
Refuted	Supported	NEI	both wrong	4	Changes one incorrect decisive label to another incorrect label.

Table 24: Frozen-validation transition patterns for AND. These aggregate audit counts are not used by the router and do not support post-hoc threshold changes. They make the main trade-off explicit: AND repairs direct decisive errors on gold NEI examples, but can over-defer correct Supported/Refuted predictions.

Transition and ID	Claim (or abridged cue)	Direct → AND	Retrieved-evidence reading and interpretation
Helpful: gold NEI; mocheq-test-claim-100409	“Over a million people [were] apprehended at the border this year, meaning probably another 2 or 3 million have come in illegally.”	Refuted → NEI	One passage confirms that more than one million people were apprehended in FY2021; another says the extrapolation to millions of undetected crossings is not well supported because the apprehension share is higher than assumed. The evidence therefore does not resolve the compound claim. AND’s deferral is correct, whereas the direct expert is overconfidently decisive.
Harmful: gold Supported; mocheq-test-claim-101393	“The Florida 45th Congressional District is among the 10 districts in the nation receiving the least federal funding.”	Supported → NEI	The retrieved report explicitly places Florida’s 45th district among the ten lowest-funded districts. The direct prediction is correct, but AND defers it. This is an instance of conservative over-deferral: a rationale-trained posterior can trigger NEI despite decisive retrieved support.
Both wrong: gold Supported; mocheq-test-claim-10007	Rick Moranis claim	Refuted → NEI	The first retrieved passage concerns Moranis and his children, but the second is an unrelated Biden quotation. The direct expert is already wrong; AND changes the output to another wrong label rather than repairing a retrieval mismatch.
Both wrong: gold Refuted; mocheq-test-claim-100909	Pre-existing-condition claim	Supported → NEI	The retrieved material is about a different Biden statement and does not resolve the claim’s speaker and proposition. AND avoids the direct expert’s unsupported decisive prediction, but the available evidence is still insufficient to recover the gold Refuted label.

Table 25: Representative raw-official-P1 routing cases. Selection is deterministic: the lexicographically first claim in each predeclared transition stratum, with at most two retrieved passages inspected per case. The two latter rows make explicit that a deferral can be non-repairing when both outputs are wrong. Evidence readings are concise author paraphrases, not generated rationales; neither test labels nor these readings are available to AND. Full claim text and passages for all twelve frozen case cards are retained in the supplementary artifact.

Track	Teacher	Acc.	MF1	F1-S	F1-R	F1-NEI	Pairwise P
Raw P1	Qwen2.5-3B	.56839	.54829	.54705	.65437	.44346	–
Raw P1	Qwen2.5-7B	.55979	.53986	.52416	.65332	.44209	.1166 (Q7>Q3)
Raw P1	Mistral-7B	.56061	.53965	.52082	.65400	.44413	.4817 (M>Q7)
Strict P1	Qwen2.5-3B	.56738	.54660	.54633	.65466	.43881	–
Strict P1	Qwen2.5-7B	.55957	.53906	.52424	.65421	.43875	.1365 (Q7>Q3)
Strict P1	Mistral-7B	.55957	.53808	.51840	.65427	.44156	.4454 (M>Q7)

Table 26: Matched teacher capacity and family ablations. For Qwen2.5 7B minus 3B, raw/strict Δ MF1 are $-.00844/-.00753$; 95% CIs are $[-.02223, .00562]/[-.02147, .00626]$. For Mistral-7B minus Qwen2.5-7B, they are $-.00021/-.00099$ with CIs $[-.01364, .01274]/[-.01376, .01183]$. Neither contrast resolves statistically.